SURFACE by MONARO

V1.0

USER MANUAL

A shaping processor for drums: noise, saturation, bit crushing and transient shaping in a reorderable chain, framed by two equalisers and level-matched output.



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Key combinations are written for macOS first. On Windows, `Cmd` is `Ctrl` and `Option` is `Alt`.
Names of on-screen controls, menus and buttons are set like this.

1 Introduction

About Surface

Most of what makes a hi-hat sit in a mix happens in the first few milliseconds and in the texture of its tail: how hard the stick lands, how much air and grit there is around it, how quickly it dies away.

Surface has four shaping modules, each changing one aspect of a hit, and matches the output level to the input so that a change in shape is not heard as a change in level. It works on any percussive material (snares, shakers, room tracks, drum buses), but its defaults, ranges and detection are tuned for hats and cymbals.

What is in it

- **Noise**
  White, pink or cassette hiss that follows each hit's envelope, for air and texture.
- **Saturation**
  Tube (even harmonics first) or Tape (odd harmonics first), oversampled with linear-phase filters.
- **Crusher**
  Sample-rate and bit-depth reduction from one Amount control, with a tone filter.
- **Transient**
  Per-hit attack and decay shaping with look-ahead, level-independent detection.

EQ In and EQ Out
Up to five bands each (cuts, shelves, bells, tilt) over a live analyser.

Reorderable chain
Drag the four modules into any order.

Auto Gain
Matches output loudness to input with one steady gain, for fair before / after comparisons.

Intensity
One control that scales every module towards neutral at once.

Workflow
Live displays for every module, presets, two themes and five interface sizes.

System requirements

PLATFORM	FORMATS	REQUIREMENTS
macOS	VST3, Audio Unit	macOS 11 Big Sur or later; Apple silicon and Intel (universal binary)
Windows	VST3	Windows 10 or 11, 64-bit host

Surface needs an internet connection once to activate (or to start the trial). After that it runs offline; see License and support. Pro Tools is not supported yet (no AAX version).

2 Quick start

Installation

macOS

1. Open `Surface_Installer.pkg`. Because the installer is not yet notarised, macOS may block it the first time: open System Settings > Privacy & Security and click Open Anyway.
2. Choose the formats you want (VST3, Audio Unit) and follow the installer. The plug-ins go to `/Library/Audio/Plug-Ins/VST3` and `/Library/Audio/Plug-Ins/Components`.
3. Restart your DAW and let it rescan plug-ins if it asks.

Windows

1. Run `Surface_Installer.exe` and confirm the administrator prompt. The installer is not yet code-signed, so Windows SmartScreen may warn the first time: click More info, then Run anyway.
2. Follow the installer. Surface is installed to `C:\Program Files\Common Files\VST3`, the standard VST3 folder every host scans.
3. Restart your DAW and let it rescan plug-ins if it asks.

To uninstall on Windows, use Settings > Apps (“Surface (MONARO)”). On macOS, delete `Surface.vst3` and `Surface.component` from the folders above.

Integrity check Release builds check their own signature; a modified copy shows “This copy of Surface has been modified”. Only download Surface from the official release page.

Activation and trial

Without a license or trial, Surface passes audio through unchanged and shows the activation sheet when you open it. **License key:** paste or type the key and click Activate; a key can be active on two computers at a time. **Free trial:** click Start free trial for seven days of the full plug-in; the status bar counts down the remaining time. To check or move your license later, open the menu (top right) and choose License….

Inserting Surface in your DAW

Surface is an insert effect, mono or stereo. Put it directly on the hi-hat (or drum) track, before any bus processing.

HOST	WHERE TO FIND IT
Logic Pro	Audio FX slot > Audio Units > MONARO > Surface
Ableton Live	Browser > Plug-Ins > VST3 (or Audio Units) > MONARO > Surface; drag onto the track
Cubase / Nuendo	Inserts slot > MONARO > Surface
Studio One	Browser > Effects > MONARO > Surface; drag onto the channel
Reaper	FX button > VST3 (or AU) > Surface (MONARO)
FL Studio	Mixer insert slot > Select > Surface; rescan in Plugin Manager if it is missing
Bitwig Studio	Device browser > MONARO > Surface

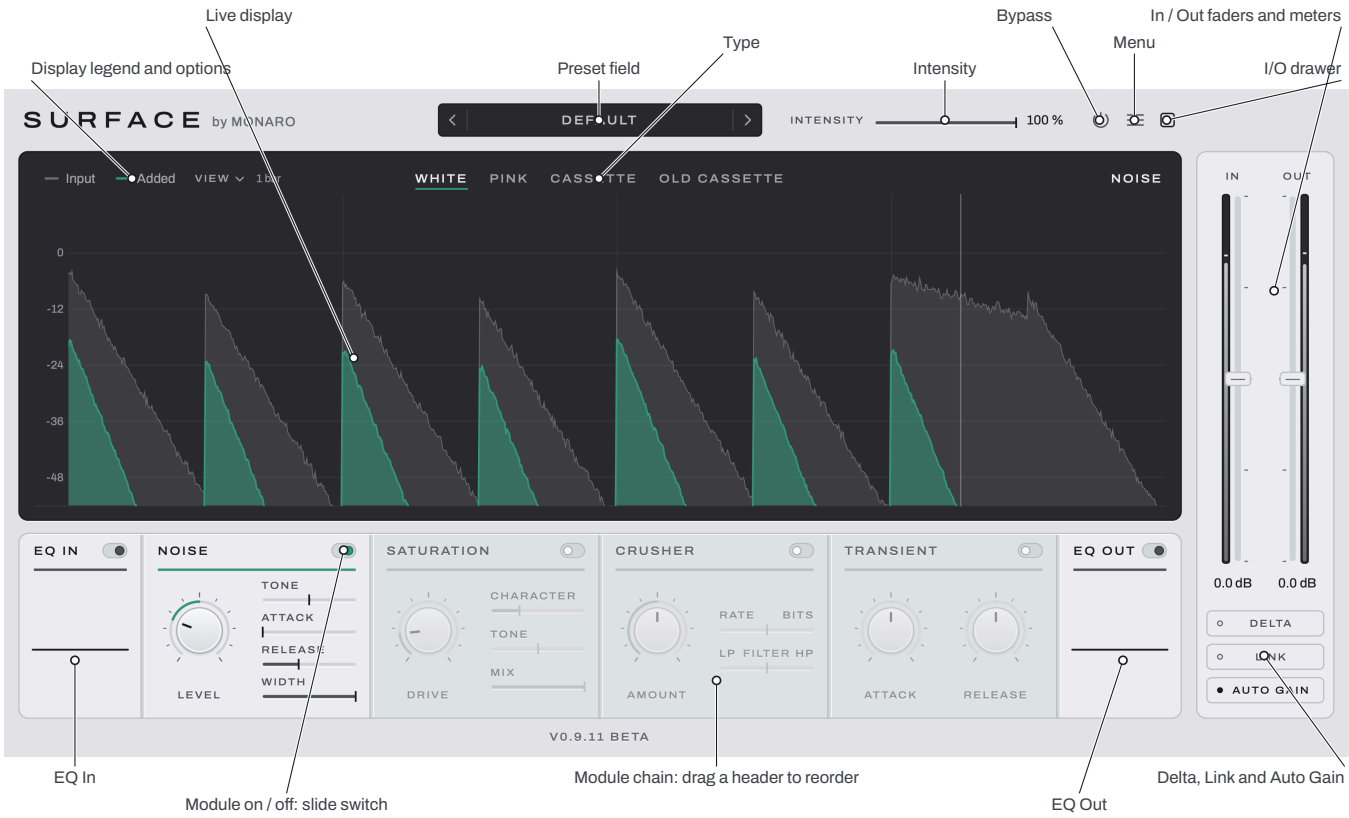
First five minutes

1. Loop a section with the hats soloed. Leave **Auto Gain** on (it is on by default): every change you make is level-matched, so louder never fools you into “better”.
2. Switch on **Transient** with the slide switch in its header. Turn Attack up for more stick, down to soften; turn Release down to tighten an open hat.
3. Switch on **Saturation**, try Tube and Tape, and raise Drive until the hats get denser without getting harsh.
4. Add **Noise** at a low Level for air between hits; pick the colour with the tabs at the top of the display.
5. Use **EQ Out** to place the result: a low cut, a little shelf at the top.
6. Compare with **Bypass** (power icon, top right), then pull **Intensity** back until the effect is just right in the mix.

3 Using Surface

Overview

The whole of Surface is on one screen: the display on top shows the selected page, the chain below holds the controls, the header holds the global actions.



Display legend and options

Click a layer to hide it; the chips set speed or resolution.

Live display

The live view of the selected page.

Preset field

Arrows step through presets; click the name for the list.

Type

The module's variant (noise colour, saturation type).

Intensity

Scales every module towards neutral.

Bypass

The power icon: bypasses the whole plug-in with a cross-fade.

Menu

Hints, interface size, theme, manual, license, About.

I/O drawer

Shows or hides the In / Out faders, Delta, Link and Auto Gain.

In / Out faders and meters

Input and output gain with peak meters.

EQ In, EQ Out

Thumbnails of the two EQs; click one to edit it in the display.

Module chain

The four modules in processing order; drag a header to reorder.

Module on / off

The slide switch at the right of each module header.

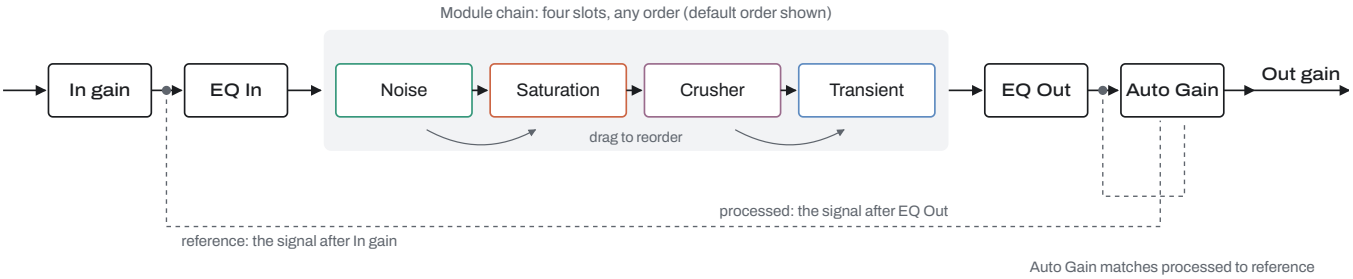
Delta, Link and Auto Gain

See Levels.

Status bar

The version, the trial countdown, short confirmations such as a saved preset; a link when a newer version is available.

Signal flow



- The four modules run in the order shown in the chain; the default order is Noise, Saturation, Crusher, Transient.
- Auto Gain compares the signal after In gain with the signal after EQ Out, so it undoes the loudness change of the processing only. Your In and Out settings stay as you set them.
- Bypass cross-fades from the processed output to the dry input, delay-compensated, so it never clicks or flams.

Knobs, sliders and faders



A control shows its value only while you move it. The same gestures work on every knob, slider and fader:

Adjust

Drag up / down (knobs, faders) or left / right (sliders), or use the mouse wheel. Hold Shift to fine-tune.

Reset to default

Cmd-click or Option-click (Windows: Ctrl- or Alt-click).

Type a value

Double-click. Enter applies, Esc cancels. Units are optional: `-3` , `-3 dB` and `-3dB` all work.

Right-click

Your DAW's own menu for that parameter (show automation, MIDI learn and so on). Where the host has none (Audio Units), a right-click types a value.

Keyboard

After tabbing to a control: arrows step 1 % (with Shift 0.1 %), Page Up / Page Down 10 %, Home resets.

In the EQ band panel, double-clicking a value opens it for typing; Option / Alt-click resets. With Show hints on (menu), the line at the bottom of the window explains the control under the mouse.

The module chain



EQ In on the left, the four modules in processing order, EQ Out on the right.

Header

Each module has its name and a slide switch at the right (on: thumb right, in the module's colour). The coloured line under the name lights while the module is on.

Select

Click the name or the module's surface: its page opens in the display.

Switch on / off

Only the switch toggles the module. Off modules sink and their controls fade, but keep their settings; switching fades in and out.

Reorder

Drag a module's header left or right; the neighbours slide aside to make room, and the module stays within the module section. Esc while dragging puts it back. The order is saved with the session and presets.

Saturating after the noise colours the noise too; crushing before the transient shaper gives the shaper grittier edges to work on; putting Transient first shapes the dry hit before anything else thickens it.

Live displays

Every page has its own live display, fed by the audio while the editor is open (the displays stop when the editor is closed). Click a legend entry (Input, Added, Output, Pre, Post) to hide or show that layer.

Level displays (Noise, Transient)

The input is a grey body. What the module adds is filled in its colour, what it removes is hatched, and the edge of the result is drawn in the same colour. VIEW switches between Static (bars standing still, in time with the song; the wheel shows 1/4, 1/2, 1, 2 or 4 bars) and Dynamic (scrolling; the wheel or a pinch zooms 0.25 to 16 s). Option / Alt + wheel zooms level, double-click resets. The view you chose last opens next time, in any session (it is not part of a preset).

Page title and kind

The page title sits at the right of the display's top row, dimmed while the module is off. Noise and Saturation choose their kind there (White / Pink / Cassette / Old Cassette; Tube / Tape) with tabs centred in that row: click, or use the mouse wheel over them, one kind per notch.

Spectrum displays (Saturation, Crusher)

The input in grey and, in colour, what the module added. Speed sets how quickly it follows, Res its frequency resolution. Drag the frequency scale to zoom and pan.

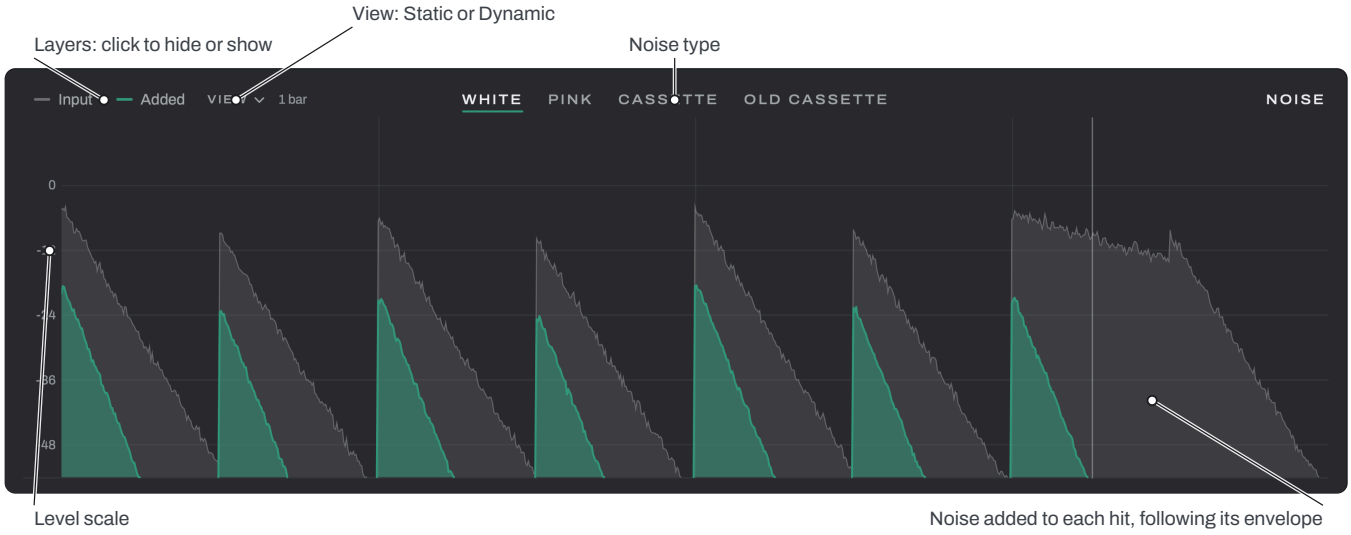
EQ In / Out

The curve over a pre / post analyser; see EQ In and EQ Out.

4 Modules

Noise

Adds noise that rises and falls with every hit. An envelope follower tracks the input and shapes the noise, so Level is relative to the hit.



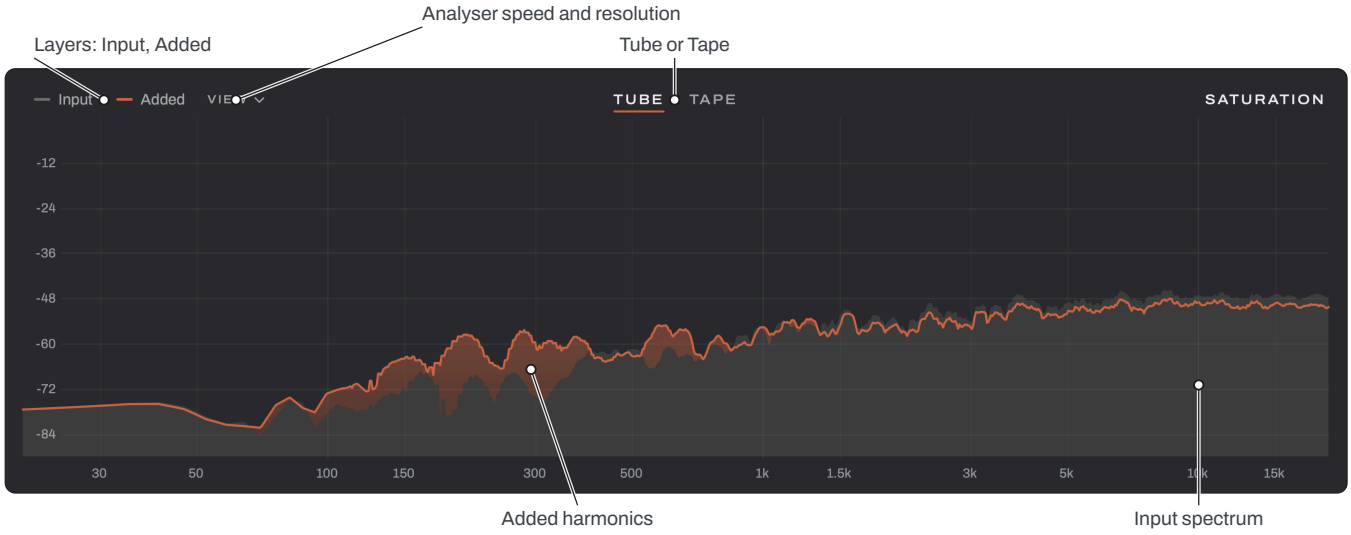


CONTROL	RANGE	DEFAULT	
Type	White, Pink, Cassette, Old Cassette	White	Colour of the noise (tabs at the top of the display). Cassette is the steady hiss of a blank Type I tape; Old Cassette a fifty-year-old tape, fuller in the lows with a little more air on top. Both are real recordings, cleaned of clicks and hum.
Level	−80 … +18 dB	−12 dB	Noise level relative to the hit. At 0 dB (knob centre) the noise is as loud as the hit.
Tone	−12 … +12 dB	0 dB	Tilts the noise around 800 Hz: left darker, right brighter. Only the colour changes, the loudness stays.
Attack	0.1 … 50 ms	1.0 ms	How fast the noise comes in. Longer values round off its start, so the hat's own attack comes through first and the noise follows it.
Release	−100 … +100 %	0 %	The noise follows the hat's level. At 0 it lasts exactly as long as the hat; to the left it is shorter, to the right it rings on past the hat, up to one second, and fades out completely by then.
Width	0 … 100 %	50 %	Stereo spread, from mono to fully decorrelated. The level stays the same at any width.

Tip A low Release with a fast Attack adds “sizzle” glued to each hit; a high Release with Cassette noise builds a lo-fi bed under a programmed pattern.

Saturation

Adds harmonics and rounds peaks. Quiet parts pass through unchanged at any Drive, so Saturation thickens the hit without changing the tonal balance of the low-level signal. The display shows the input spectrum in grey and, in orange, only what Saturation added: raise Drive and the orange layer grows; change Character and its harmonic balance shifts.





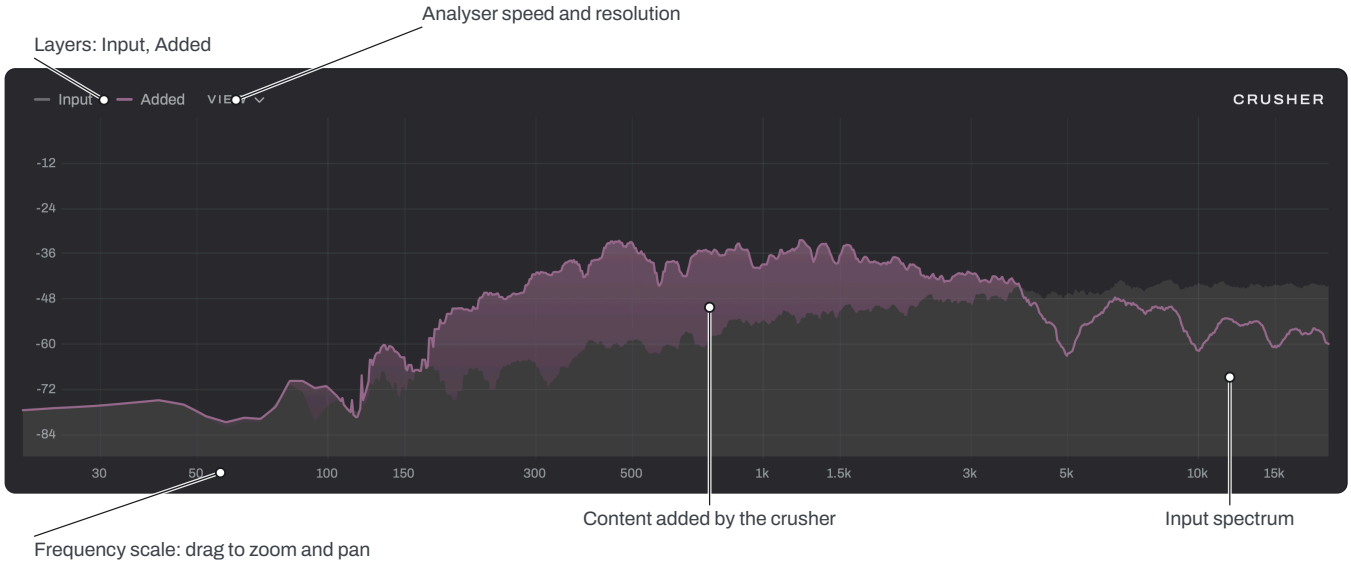
CONTROL	RANGE	DEFAULT	
Type	Tube, Tape	Tube	**Tube**: even harmonics lead (2nd above 3rd); warm, full, slightly lopsided. **Tape**: odd harmonics lead (3rd strongest) with a smaller 2nd and a rich upper series; denser and brighter.
Drive	0 … 36 dB	6 dB	How hard the signal hits the curve. Higher values add more harmonics.
Character	0 … 100 %	30 %	**Tube**: amount of asymmetry, i.e. how much 2nd harmonic. **Tape**: from new tape (clean until it saturates, firm, odd harmonics) to old tape (bends earlier and softer, warmer 2nd; the top of a loud hit saturates first and loses its edge, a quiet one passes).
Tone	−12 … +12 dB	0 dB	Which part of the hat saturates: up drives the body and keeps the top open and bright, down drives the top so it softens first. Only the saturation moves; the clean balance stays.
Mix	0 … 100 %	100 %	Blend of saturated and clean signal. Both paths are time-aligned, so blending never combs.

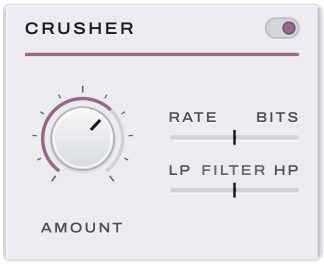
The module runs at twice the host sample rate with linear-phase filters and antialiased curves: less aliasing on the top octave and no phase shift. Below about 200 Hz only the clean signal passes, so the low end of a drum bus stays tight while the top saturates.

Level A hat keeps its loudness at any Drive, Character and Tone: the peaks are held and the body comes up, as with driven hardware. Auto Gain matches the rest.

Crusher

Digital degradation: sample-rate reduction (aliasing, metallic “ring”) and bit-depth reduction (grain, quantisation noise). Amount sets how deep both go; Rate / Bits sets the balance between them. The purple layer on the display is what the reduction added to the input.





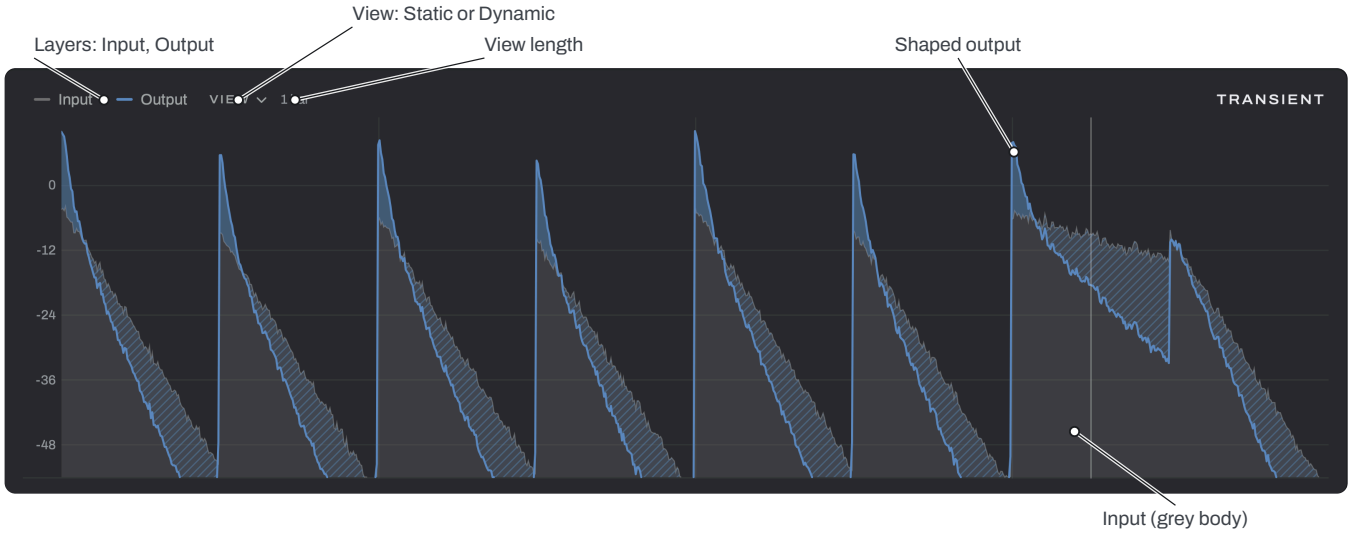
CONTROL	RANGE	DEFAULT	
Amount	0 ... 100 %	50 %	Overall depth of the reduction.
Rate / Bits	Rate 100 ... Both ... Bits 100	Both	Balance between the two: left only reduces the sample rate, right only the bit depth, centre does both fully.
Filter	LP 100 ... Off ... HP 100	Off	One-knob tone filter after the crusher, a gentle 6 dB/oct slope: left darkens (low-pass), right thins (high-pass).

The low end below 300 Hz stays clean (a phase-matched split), so crushing a full drum loop keeps its weight.

By design Crusher runs at the host rate on purpose: the aliasing is the sound.

Transient

Shapes each hit individually. Surface detects every onset independently of level, so quiet ghost notes are treated like accents, and applies a gain envelope to it. On the display the grey body is the input, what Transient adds is filled in blue, what it removes is hatched: more Attack fills the start of each hit, a negative Release hatches its tail.





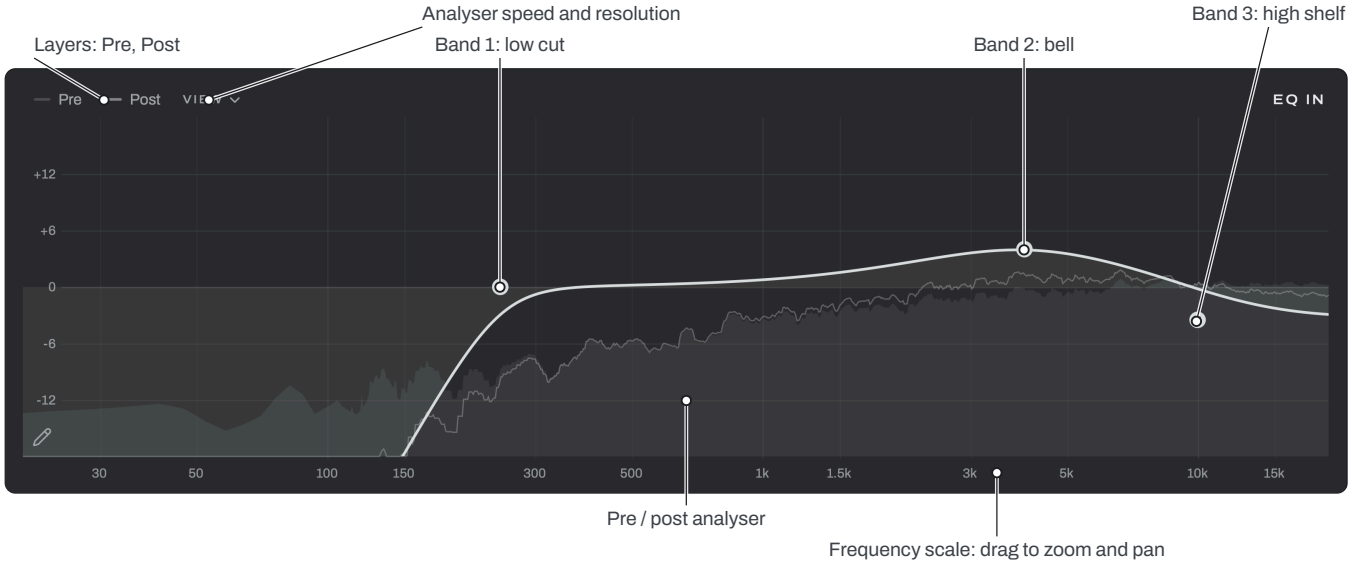
CONTROL	RANGE	DEFAULT	
Attack	−100 … +100 %	0 %	Positive: more punch on each hit. Negative: softer hits.
Release	−100 … +100 %	0 %	Positive: the hit rings longer. Negative: shorter, tighter tails. It adapts to the length of the hits, so it feels the same on a closed and an open hat.

A constant 0.5 ms look-ahead lets it shape even the very first edge of the hit. Rolls and rattles do not retrigger the shaper, and both channels are linked, so the stereo image never shifts.

Timing The look-ahead latency is the same whether the module is on or off, so switching it never moves the timing.

5 EQ In and EQ Out

Two equalisers frame the modules. **EQ In** prepares the signal before it is shaped (for example, cut the low end so Saturation does not work on bleed); **EQ Out** places the result in the mix. Each has up to five bands and its own on / off switch; click the thumbnail at either end of the chain to edit it in the display.



Q follows the convention of most equalisers: Q 1 is a plain (Butterworth) cut and a bell of about two octaves, so the same numbers give the same curves as in other EQs. The bands follow their analog prototypes all the way up to the top of the spectrum, without the high-frequency squeeze of a standard digital EQ, and with minimum phase. The analyser shows the signal before (Pre) and after (Post) the EQ; Speed and Res set its ballistics and resolution.

Band types

TYPE	SETTINGS
Low Cut / High Cut	Frequency; slope 6, 12, 18, 24, 30, 36, 48, 72 or 96 dB/oct (default 12); resonance (the height of its point)
Low Shelf / High Shelf	Frequency, gain ±18 dB, Q
Bell	Frequency, gain ±18 dB, Q 0.1 … 18
Tilt	Centre frequency and gain: tilts the whole spectrum around it

Creating and editing bands

Create

Click an empty spot: at the very edges a cut (below 50 Hz, above 15 kHz), next to them a shelf (up to 120 Hz, from 7 kHz), elsewhere a bell. Hovering previews the band. Click-and-drag on the curve always creates a band and moves it, even with a band selected. Double-clicking the background also adds a band.

Move

Drag a point: frequency and gain. Shift-drag moves it more slowly. Option-drag locks the drag to the axis it starts on. Cmd-drag changes frequency only.

Q and slope

Mouse wheel over a point: Q. On a cut, one notch is one slope step (up steeper, down gentler). Cmd + wheel changes gain, Shift + wheel makes finer steps.

Select

Click a point; Cmd-click adds; Shift-click selects a range; drag on the background to lasso; Cmd A selects all, Esc none.

Bypass a band

Option-click its point.

Change shape

Cmd Option-click cycles the type; Option Shift-click cycles the slope.

Reset and exact values

Double-click a point to reset its gain to 0 dB (on a cut, its Q). Type exact values in the band panel.

Context menu

Right-click a point: bypass, shape, copy, delete. Right-click the background: paste.

Delete and nudge

Select and press Delete; arrow keys nudge the selection.

Zoom

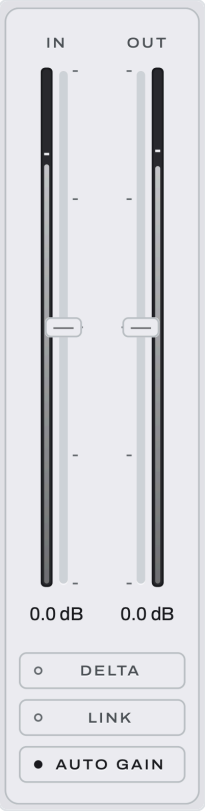
Drag the frequency scale up / down to zoom, left / right to pan; pinch or Cmd + wheel over the background; Option + wheel zooms the dB scale; double-click the frequency scale resets the view.

Draw mode

The pencil button at the bottom left of the display switches to draw mode. Sketch a curve and Surface fits bands to it; they replace the bands of that EQ. A part drawn down to −15 dB or lower becomes a cut. The mode ends when you release the mouse; Esc leaves it.

On Windows use Ctrl for Cmd and Alt for Option.

6 Levels: I/O, Auto Gain, Intensity



I/O drawer

The I/O button (top right) slides out two faders with peak meters, and the Delta, Link and Auto Gain switches. The meters show peak level in dBFS with a short peak hold; the hold mark turns orange above 0 dBFS.

CONTROL	RANGE	DEFAULT	
In	−24 … +24 dB	0 dB	Gain into the processing. Driving the input harder makes Saturation and Crusher work harder and changes what the Noise envelope follows.
Out	−24 … +24 dB	0 dB	Final output gain.
Delta	On / Off	Off	Plays only what Surface changes: the output minus the latency-aligned input (after In gain), level-matched when Auto Gain is on. Out gain still applies. Not stored in presets.
Link	On / Off	Off	With Link on, the output gain is Out – In: you can push the input into the modules, and moving or automating In does not change the output level. The Out fader shows the gain that reaches the output.
Auto Gain	On / Off	On	Matches output loudness to input loudness.

Auto Gain

Auto Gain applies one gain to the whole signal: a hit recorded 15 dB under the next stays 15 dB under it. The gain follows knob moves and automation within about 20 ms. A slow correction is applied at each onset, never inside a hit, so it cannot pump or reshape transients. On material without clear hits the correction updates every 1.5 seconds. For a fresh instance a safety guard keeps the first hits from jumping out while the first measurement is taken. Auto Gain lifts the level by at most 12 dB: when the processing all but silences the hits (heavy Crusher on very quiet material, a steep high cut) the output stays quieter rather than jumping.

When to turn it off When you want Transient to make the track louder on purpose, or when the next plug-in in the chain depends on the absolute level.

Intensity

Intensity (header) scales every module from neutral (0 %) to exactly as set (100 %, default). It is not a dry/wet mix: only each module's main knob is reduced (Noise Level, Saturation Drive, Crusher Amount, Transient Attack and Release), so the character, mix and filter settings stay as you set them and there is no phase combing between a dry and a processed copy. Below 10 % Saturation eases fully to neutral. Automate it to change the effect between sections.

7 Presets and bypass

Presets

Browse

The arrows next to the preset name step to the previous / next preset.

Preset menu
Click the name: Init (all defaults), the presets grouped by folder (the factory presets are in MONARO), Save As..., and Options.

Default
Options > Save as Default makes the current settings the startup state for new instances (badged in the menu); Reset Default returns to the factory start.

Import
Options > Import accepts `.surfacepreset` files or a `.zip` of them; folders inside the archive are recreated.

Presets store every setting including the module order. They are plain files (`*.surfacepreset`), so you can copy, share and organise them in sub-folders. They live in `~/Library/Audio/Presets/MONARO/Surface` (macOS) and `Documents\MONARO\Surface` (Windows).

Bypass

The power icon bypasses the whole plug-in with a short cross-fade. The dry path is delay-compensated, so bypassing never shifts timing. While bypassed, the window darkens and says BYPASSED. Bypass is also available to your host as a parameter.

8 Interface settings

All in the menu, top right:

Show hints	The line at the bottom of the window says what the control under the mouse does.
Interface size	50, 75, 100, 125 or 150 %. Remembered with the session.
Theme	Graphite or Chalk.
User manual	Opens this document.
License...	Shows the license holder; activate or deactivate this computer.
About	Version and copyright.

9 Technical notes

Latency

Surface reports a constant latency to the host, which compensates it automatically.

SOURCE	LATENCY	REASON
Saturation	97 samples	Linear-phase oversampling filters
Transient	0.5 ms	Look-ahead
Total at 44.1 kHz	≈ 2.7 ms	
Total at 48 kHz	≈ 2.5 ms	

The latency is the same whether modules are on or off and in any order, so switching and reordering never move the timing. When you monitor live through Surface, you hear this latency.

Transparency

With every module off, Surface is bit-transparent apart from the reported delay. A module switched on at neutral settings does not change the frequency response or phase of a quiet signal (within ±0.05 dB up to 21 kHz at 44.1 kHz). There is no global oversampling and no dry/wet mixing at the plug-in level, both deliberate choices to avoid comb filtering.

10 License and support

Your license

Two computers
: One license can be active on two computers at a time. To move it, open License... on the old computer and choose Deactivate, then activate on the new one.

Offline use
: After activation Surface works offline. It renews its activation in the background when online; it keeps working for at least 45 days without a connection.

Trial
: The trial runs for seven days on one computer and is not tied to a key. When it ends, Surface passes audio through unchanged until you activate a license.

Troubleshooting

PROBLEM	SOLUTION
Surface is not in my plug-in list	Rescan plug-ins in your host. On macOS check that the formats were selected in the installer. Make sure the host scans the standard VST3 folder.
“This copy has been modified”	Run the installer from the official release page again.
Activation fails	Check the internet connection and the key (copy and paste it). “Limit reached” means two computers are already active: deactivate one.
Level jumps when comparing	Make sure Auto Gain is on; it follows a change within about 20 ms.

Support

For help, bug reports and license questions, contact MONARO through the channel you received your license from. Please include your operating system, host and version, and the Surface version shown in About.

11 Acknowledgements

Surface is designed and developed by MONARO. Thanks to everyone who tested the betas.

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